

Sequential Data Driven Optimisation of Formula 1 Race Strategy: A Literature Survey

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Abstract—Formula 1 race strategy requires sequential decision-making under uncertainty, where pit stop timing, tyre degradation, safety car probability, traffic dynamics, and competitor behaviour jointly determine race outcomes. Existing research has progressed from regression-based degradation modelling to deterministic and stochastic optimisation, game-theoretic interaction modelling, and machine learning prediction frameworks. However, these approaches often address isolated components of the strategy problem and rarely integrate prediction, uncertainty modelling, and prescriptive optimisation within a unified architecture.

This study proposes a modular data-driven decision framework that combines probabilistic lap-time modelling, safety car event prediction, opponent behaviour modelling, Monte Carlo race simulation, and offline reinforcement learning. By treating race strategy as a sequential multi-agent decision process rather than a static optimisation task, the framework enables uncertainty-aware policy learning. The proposed system aims to bridge the gap between predictive analytics and real-time strategic decision intelligence in Formula 1.

Index Terms—Formula 1, Pit Stop Strategy, Sequential Decision Making, Stochastic Simulation, Reinforcement Learning, Machine Learning

I. INTRODUCTION

Formula 1 is one of the most data-heavy and strategically difficult sports in the modern world. The results of a race are not based only on how well the cars perform, but also on a series of strategic choices taken when there is ambiguity. During the race, the timing of pit stops, the choice of tire compound, the way tires wear down, the use of safety cars, the positioning of traffic, and the responses of competitors all change. Every choice affects the future, which means that strategy is always changing and never stays the same.

Early analytical methods looked at when to make a pit stop as a scheduling or optimization problem with simple assumptions. As the sport progressed, researchers used stochastic modeling, game-theoretic frameworks, and machine learning techniques to address uncertainty and competitive dynamics. Even while their contributions made things more realistic and better at predicting what would happen, most of the models we have now only look at one part of the strategy problem at a time. Most prediction models don't have prescriptive optimization layers, and most optimization frameworks don't include probabilistic forecasting or adaptive learning methods.

In real life, Formula 1 teams use scenario modeling, probabilistic reasoning, and competitive anticipation to figure out their strategies. This means that race strategy should be thought of as a multi-agent sequential decision process with many possible outcomes, not as a single-step optimization assignment.

Driven by this deficiency, this study conceptualizes race strategy as a cohesive decision intelligence system. The proposed framework includes probabilistic lap-time modeling, stochastic event prediction, estimating how an opponent would act, Monte Carlo race simulation, and reinforcement-based policy learning. The system's goal is to go beyond static pit stop modeling and toward uncertainty-aware strategic policy development by combining prediction and optimization in a modular framework.

II. LITERATURE SURVEY

Carrasco, Heine, and Thraves [1] approached pit-stop optimisation in Formula 1 as a dynamic programming problem, aiming to minimise total race time while considering tyre degradation. Their model demonstrates how sequential decision-making can determine optimal pit timing and offers a clear analytical framework. However, it assumes deterministic race conditions and fixed tyre wear patterns. Uncertainties such as safety car events, changing weather, or competitors' strategies are not included. As a result, the model provides a structured solution but cannot adapt to unpredictable race scenarios or learn from data in a probabilistic manner.

Heine and Thraves [2] extended this work by refining tyre degradation models and improving computational efficiency within a broader dynamic programming framework. These improvements make the model more realistic and allow faster strategy evaluation. Despite these advances, the method still relies on known system dynamics and does not account for stochastic events or opponents' behaviour. Therefore, it remains focused on deterministic optimisation and does not explore adaptive or reinforcement-based strategies under uncertainty.

Greasley, Panchal, and Samvedi [3] introduced a digital twin framework that integrates simulation, optimisation, and machine learning to support race strategy decisions. Their system

combines predictive models with simulation to assess different tactical options in a virtual environment. This approach highlights the benefits of simulation-driven analytics, but it does not formalise strategy selection as a sequential learning problem. Probabilistic risk modelling, opponent behaviour prediction, and reinforcement learning are not incorporated, so the system evaluates strategies without actively learning optimal policies.

Paparusso, Riani, Ruggeri, and Braghin [4] developed a stochastic optimisation model for hybrid race vehicles that considers competitor strategies and lap time uncertainty. This method accounts for multi-agent interactions and introduces stochastic elements into strategy optimisation. However, it focuses on trajectory optimisation and does not integrate latent behavioural modelling or reinforcement-based decision-making. The system optimises expected outcomes but does not learn sequential policies from past data.

Duhr et al. [5] investigated energy allocation strategies for hybrid-electric Formula 1 engines using optimal control to minimise race time. The study focuses on continuous energy management under power unit constraints rather than discrete strategic decisions like pit stops. Probabilistic events such as safety car deployment or opponent actions are not included. Therefore, the contribution remains within deterministic optimisation and does not address adaptive policy learning.

Quiroga [6] explored a data-driven approach to pit-stop optimisation using historical race data and predictive modelling. The study evaluates potential strategies through scenario analysis and statistical learning. While it provides empirical insights into pit timing, it is primarily predictive rather than prescriptive. Safety car hazards, opponent behaviour, and stochastic exploration of race futures are not systematically included, limiting the framework's ability to generate adaptive, uncertainty-aware strategies.

Rao [7] used ensemble learning to predict pit-stop decisions through supervised classification. While effective at anticipating driver behaviour based on race context, the method treats strategy as a static prediction problem. It does not simulate future race scenarios or optimise sequential actions, focusing on historical behaviour rather than learning policies that adapt to dynamic conditions.

Patil et al. [8] analysed Formula 1 outcomes using machine learning models to predict finishing positions. This work demonstrates the predictive power of statistical learning but is oriented toward outcome prediction rather than strategic intervention during the race. Sequential decision-making and probabilistic policy optimisation are not part of the framework. While predictive analytics are improved, the study does not provide actionable guidance for adaptive race strategy under uncertainty.

Recent advances in reinforcement learning (RL) and multi-agent systems present promising avenues for addressing the limitations identified in prior studies. By framing pit-stop and race strategy decisions as sequential decision-making problems under uncertainty, RL algorithms can actively learn optimal policies from historical race data and simulated scenarios.

This allows models not only to consider tyre degradation and energy management but also to dynamically adapt to stochastic events such as safety car deployments, weather changes, or unexpected competitor maneuvers. Integrating these approaches with predictive components from machine learning [6]–[8] can provide a hybrid framework where data-driven insights guide exploration while reinforcement-based optimisation refines tactical decision-making in real-time.

Furthermore, simulation-based digital twin environments [3] can serve as a testbed for such adaptive strategies. By embedding probabilistic models of opponent behaviour and race contingencies, these systems enable agents to evaluate multiple policy trajectories before deployment in actual races. Combining stochastic modelling techniques [4] with reinforcement learning allows the system to balance risk and reward, optimising expected performance while maintaining robustness against unpredictable events. Such a methodology overcomes the deterministic assumptions in earlier optimisation-focused works [1], [2], [5] and provides a more realistic, operationally relevant framework for strategy planning in Formula 1 contexts.

Recent studies in hybrid optimisation approaches suggest the potential for cross-disciplinary integration. For instance, combining ensemble predictive models [7] with adaptive RL policies can improve initial strategy estimates while enabling continual learning from unfolding race conditions. This dual approach ensures that both historical patterns and live feedback are leveraged, resulting in strategies that are both evidence-based and responsive. Future research could focus on developing end-to-end systems that merge simulation, predictive analytics, and reinforcement learning to create fully adaptive, multi-agent-aware race strategy platforms, bridging the gap between theory and practical race-day decision-making.

Using real-time telemetry and sensor data from the cars can improve adaptive race strategies. By feeding live information such as tyre temperature, track grip, and fuel levels into predictive models, strategies can be updated continuously during the race [1], [2], [6]. This allows pit stops and energy use to be adjusted according to current conditions rather than relying solely on historical patterns. With this feedback, reinforcement learning models can refine decision-making as the race progresses, improving performance in uncertain situations.

Multi-agent reinforcement learning (MARL) can also help model interactions between different teams and drivers. Unlike single-agent approaches, MARL considers how each competitor may act and react, providing a more realistic simulation of race dynamics [4], [7].

Finally, when combined with stochastic modelling and predictive analytics [3], this allows the system to generate strategies that adapt to both changing conditions and the actions of opponents. Such integration addresses a major limitation in previous work, offering a framework that is flexible, responsive, and capable of sequential decision-making under uncertainty.

TABLE I: Comparative Analysis of Existing Formula 1 Strategy Optimisation Approaches

Paper	Aim	Methods Used	Research Gap
Carrasco Heine and Thraves [1]	To determine optimal pit stop timing for minimising total race time under tyre degradation effects	Dynamic programming based deterministic optimisation	Does not account for uncertainty in safety car events or competitor behaviour
Heine and Thraves [2]	To improve pit stop strategy computation through refined tyre degradation modelling	Enhanced dynamic programming framework with improved degradation representation	Ignores probabilistic race events and adaptive opponent interaction
Greasley, Panchal, and Samvedi [3]	To evaluate race strategy options using predictive modelling in simulated environments	Digital twin framework integrating machine learning with simulation	Lacks sequential policy learning and probabilistic hazard modelling
Paparusso, Riani, Ruggeri, and Braghin [4]	To optimise race trajectory by considering lap time uncertainty and competitor interaction	Competitor aware stochastic optimisation modelling	Does not incorporate reinforcement learning for adaptive decision making
Duhr et al. [5]	To minimise race completion time through efficient hybrid energy deployment strategies	Optimal control based continuous energy allocation	Focuses on energy management rather than pit stop strategy decisions
Quiroga [6]	To analyse pit stop timing strategies using historical race telemetry data	Data driven predictive modelling with scenario based analysis	Strategy treated as prediction rather than sequential optimisation problem
Rao [7]	To anticipate pit stop decisions based on contextual race features	Supervised ensemble learning classification approach	Does not simulate future race scenarios for decision optimisation
Patil et al. [8]	To predict race finishing outcomes from multi source race statistics	Machine learning based outcome prediction models	Lacks sequential strategy optimisation for real time decision support

A. Summary

The objectives and methodological approaches adopted in existing research on Formula 1 race strategy optimisation are summarised in Table I. A large proportion of the literature is focused on the improvement of pit stop timing and overall race performance, which is achieved by modelling tyre degradation and stint behaviour through deterministic optimisation techniques such as dynamic programming [1], [2]. Known system dynamics and stable race conditions are typically assumed in these approaches.

In subsequent studies, uncertainty is incorporated through stochastic optimisation or simulation-driven digital twin environments, allowing variability in lap time behaviour and competitor interaction to be captured [3], [4]. However, predefined strategies are primarily evaluated by these methods, rather than adaptive policies being learned over sequential decision stages.

In more recent work, data-driven techniques are adopted, and historical telemetry along with contextual race features are utilised to predict pit stop decisions or race outcomes [6]–[8]. Although improvements in prediction accuracy are reported, the strategy problem is generally treated as a static forecasting task rather than being approached as a dynamic optimisation problem.

Across the reviewed studies, limited emphasis is placed on the modelling of race strategy as a sequential decision-making process under uncertainty. In particular, probabilistic modelling of stochastic race disruptions, such as safety car deployment, adaptive opponent behaviour, and real-time policy learning, remains insufficiently explored.

III. CONCLUSION

Three main methodological approaches are primarily adopted in the existing literature on Formula 1 strategy. In early work, deterministic dynamic programming is applied

to compute optimal pit timing under predefined degradation dynamics and known race conditions [1], [2]. In subsequent studies, stochastic elements and competitor awareness or optimal control formulations are incorporated, yet the approaches remain largely trajectory-based and model-driven rather than policy-oriented [4], [5]. More recent contributions employ machine learning and simulation for pit-stop or race outcome prediction, with a focus on forecasting rather than on decision optimisation [3], [6]–[8].

The proposed framework represents an advancement of this body of work by formulating race strategy as a sequential decision problem under uncertainty. Probabilistic lap time modelling, hazard modelling for race disruptions, opponent behaviour inference, Monte Carlo simulation, and offline reinforcement learning are integrated to enable the learning of an optimal policy. Through this integration, the work is positioned beyond deterministic optimisation and predictive modelling toward prescriptive, policy-based strategy optimisation.

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